

When using a uniform noise schedule, tokens are often unmasked in positions that lack positional context. I tried to add an inductive left-to-right bias into the diffusion process and it seems that the uniform noise schedule is a special case of a wider class of Normal noise schedules.

Gaussian Noise Schedule

Let L be the sequence length and $t \in [0, 1]$ a continuous timestep, where $t = 0$ corresponds to clean data and $t = 1$ to fully noised data. For hyperparameter $\sigma > 0$, define the sweep width:

$$W = L + 4\sigma \quad (1)$$

The masking probability at position $i \in \{0, \dots, L-1\}$ and timestep t is the CDF of a Gaussian whose mean sweeps monotonically from right to left across the sequence:

$$\mu(t) = (1-t)W - 2\sigma \quad (2)$$

$$p_{\text{mask},i}(t) = \Phi\left(\frac{i - \mu(t)}{\sigma}\right) \quad (3)$$

where Φ denotes the standard normal CDF. At $t = 0$, the mean $\mu(0) = L + 2\sigma$ lies far to the right of the sequence, giving $p_{\text{mask},i}(0) \approx 0$ (clean). At $t = 1$, the mean $\mu(1) = -2\sigma$ lies far to the left, giving $p_{\text{mask},i}(1) \approx 1$ (fully masked).

The probability that position i retains its clean token (unmasked signal) is:

$$\alpha_i(t) = 1 - p_{\text{mask},i}(t) \quad (4)$$

The time derivative of the masking probability, required for the ELBO (see Section), follows from the chain rule:

$$\frac{\partial p_{\text{mask},i}}{\partial t}(t) = \phi\left(\frac{i - \mu(t)}{\sigma}\right) \cdot \frac{W}{\sigma} \quad (5)$$

where $\phi = \Phi'$ is the standard normal PDF. The factor W/σ arises because $d\mu/dt = -W$, giving $\partial/\partial t (i - \mu(t))/\sigma = W/\sigma$.

We illustrate an example schedule with $L = 258$, $\sigma = 20.0$ in Figure 1.

ELBO Loss

Let $\mathbf{x}_0 \in \mathcal{V}^L$ be a sequence over vocabulary $\mathcal{V}$, and let $\mathbf{x}_t$ be the corrupted sequence at time t , where each token i is independently replaced by the mask token [M] with probability $p_{\text{mask},i}(t)$, and remains visible otherwise. Let $p_\theta(\cdot | \mathbf{x}_t)$ denote the model’s output distribution.

We use subs parameterisation from MDLM [1].

The token-averaged ELBO training objective is:

$$\mathcal{L} = \frac{1}{|\mathbf{x}_t|} \sum_{i=1}^L w_i \cdot \frac{\partial p_{\text{mask},i}/\partial t}{p_{\text{mask},i}(t)} \cdot \left(-\log \tilde{p}_\theta(x_{0,i} | \mathbf{x}_t)_i\right) \quad (6)$$

where $p_{\text{mask},i}(t)$ is clamped to $[\varepsilon, 1-\varepsilon]$ for numerical stability ($\varepsilon = 10^{-5}$). The per-token weight w_i upweights the end-of-sequence token by a factor $\lambda > 1$:

$$w_i = \begin{cases} \lambda & \text{if } x_{0,i} = [\text{EOS}] \\ 1 & \text{otherwise} \end{cases} \quad (7)$$

The ratio $(\partial p_{\text{mask},i}/\partial t) / p_{\text{mask},i}(t)$ is the standard ELBO weighting from masked diffusion [1], but here we use a position-dependent schedule instead of the uniform schedule.

Results

We ran a sweep with $T \in 50, 100, 500$ and $\sigma \in 10.0, 20.0, 40.0$. The backbone model is the transformer with a combination of rotary and sinusoidal positional encoding model. With a uniform noise schedule, this model capped at 0.955 accuracy, while with Gaussian, we managed to achieve 1.0. Full details of the model are given in Table 1.

In Figure 2, we show the plot curves for *both rules*’ accuracy on the unconditional dataset with 100 samples, i.e., the proportion of correctly denoised on inputs with $\text{max}L = 258$ mask tokens. We used 50k training epochs for each experiment.

Analysis

We see that across a constant T , as σ increases, the accuracy generally decreases. At low σ , the ‘active window’ of denoising (i.e., the slanted part of the CDF curve where the probability of a clean signal is above 0 and below 1) is only a few tokens, essentially simulating an autoregressive generation process. As σ increases, the Gaussian CDF curve flattens and approaches the uniform distribution, re-introducing the problem of tokens in the suffix being unmasked while the majority of the prefix remains masked.

Both autoregressive and diffusion processes can be modeled by controlling σ , and we should be able to prove this mathematically?

The setup with $T = 50$ achieves 1.0 accuracy but with a 5x inference speedup compared to the autoregressive generation which requires $\text{max}L = 258$ iterations.

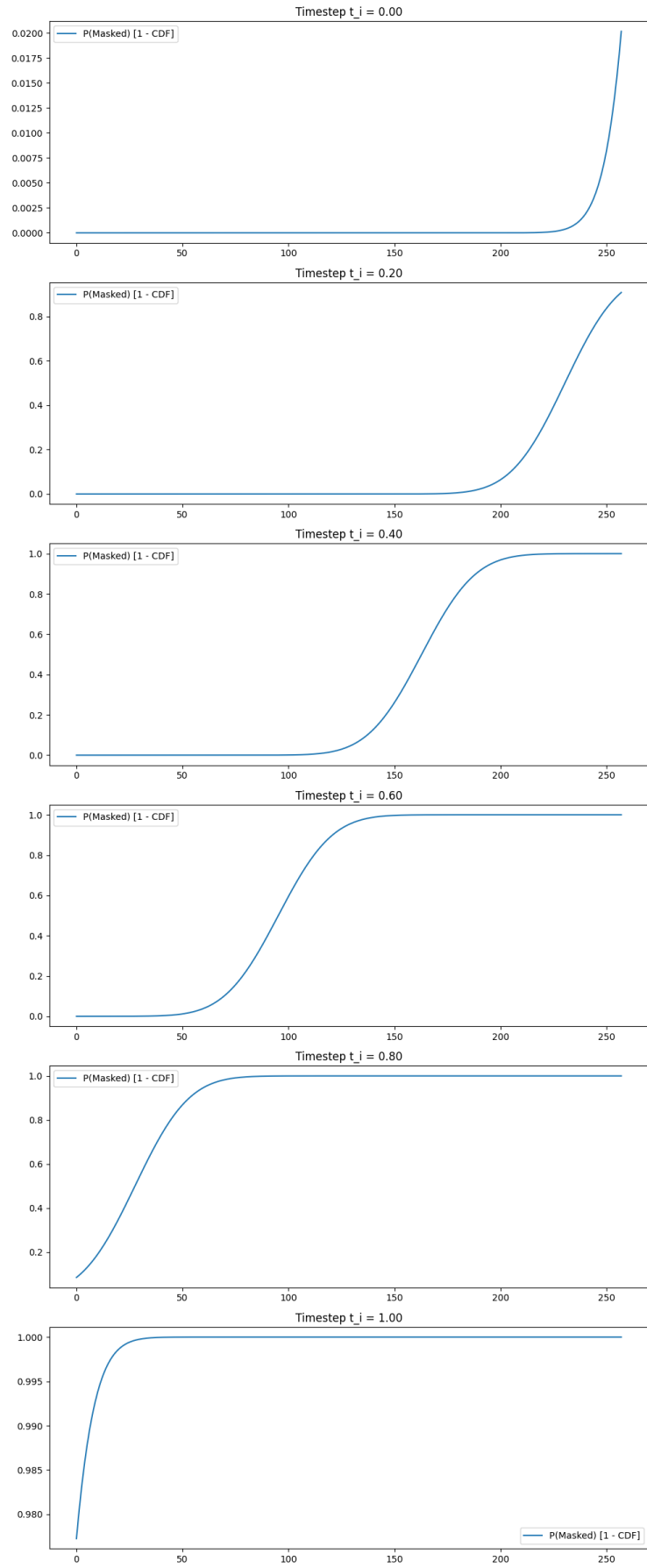


Figure 1: An example schedule with $L = 258$, $\sigma = 20.0$

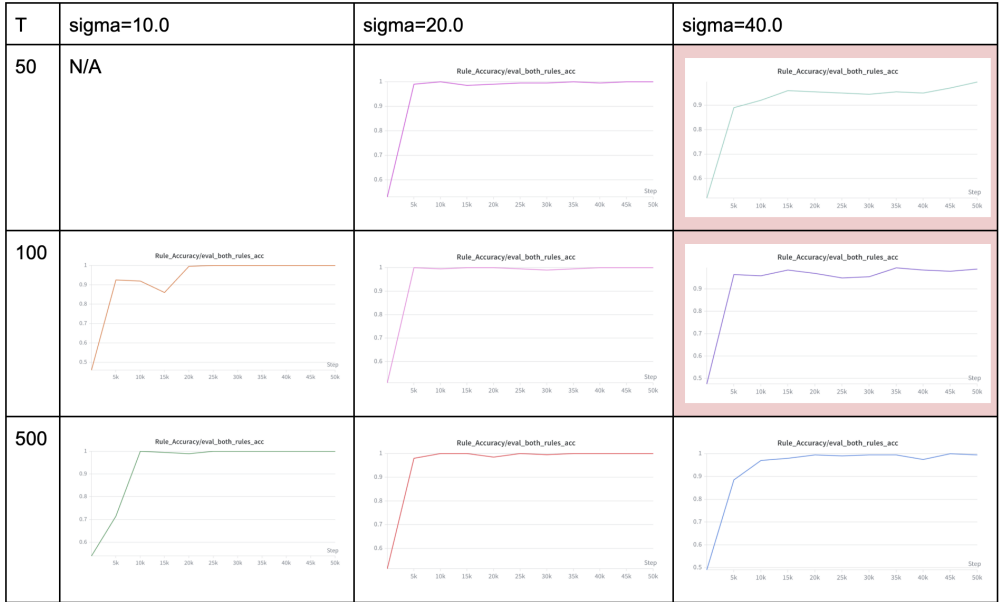


Figure 2: Accuracy plot curves for varying T and sigma parameters. Highlighted in red are the experiments where accuracy only reached 0.995; all other experiments reached 1.0.

Table 1: Hyperparameters used for training and evaluating the Gaussian discrete diffusion model on the $a^n b^n$ grammar task.

Category	Hyperparameter	Value
Data	Grammar	$a^n b^n$
	Sequence Length (L)	256
	Max Sequence Length Including SOS,EOS ($max L$)	258
	Vocabulary Size	6
	Batch Size	64
	Train/Val Split	90% / 10%
Model Architecture	Backbone	Transformer with RPE
	Hidden Dimension (d_{model})	256
	Feedforward Dimension (d_{ff})	1024
	Attention Heads (n_{head})	4
	Layers (n_{layers})	4
	Dropout	0.1
	Layer Norm ϵ	2×10^{-4}
	Sampling Margin ϵ	10^{-5}
Diffusion & Schedule	Inverse t Sampling	False
	Decoder Temperature	10^{-6} (Greedy)
	Max Epochs	50,000
Training	Optimizer	AdamW
	Peak Learning Rate	0.001
	Warmup Steps	1,000
	Learning Rate Schedule	Inverse Sqrt
	Weight Decay	0.01
	Loss Formulation	ELBO (Eq. 8)
	EOS Loss Weight (λ)	2.0

References

- [1] Subham Sekhar Sahoo, Marianne Arriola, Yair Schiff, Aaron Gokaslan, Edgar Marroquin, Justin T Chiu, Alexander Rush, and Volodymyr Kuleshov. Simple and effective masked diffusion language models. In *Advances in Neural Information Processing Systems*, volume 38, 2024. arXiv:2406.07524.